A first-order primal-dual algorithm for convex problems with applications to imaging

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Introduction

Paper: Antonin Chambolle, Thomas Pock. *A first-order primal-dual algorithm for convex problems with applications to imaging*. 2010. ⟨hal-00490826⟩

Problem:

- Variational methods are essential for ill-posed inverse imaging problems.
- While total variation regularization preserves sharp discontinuities (edges), it is difficult to minimize due to the non-smoothness of the functional.
- The authors introduce a flexible first-order primal-dual algorithm specifically designed for structured, non-smooth convex optimization problems.
- Unlike prior schemes that either required expensive linear system solves per iteration or stalled at slow subgradient rates, this method achieves an optimal $O(1/N)$ convergence rate for general non-smooth problems.
- The algorithm accelerates to $O(1/N^2)$ when either the primal or dual objective is uniformly convex, and achieves linear convergence $O(1/e^N)$ when both are uniformly convex.

Optimization Problem

$$\min_{x \in X} \; F(Kx) + G(x)$$

such that:

- X, Y: finite-dimensional real vector spaces.
- $K : X \to Y$ is continuous, linear; $\|K\| = \max\{\|Kx\| : x \in X, \|x\| \leq 1\} < \infty$.
- $G : X \to [0, \infty)$ and $F : Y \to [0, \infty)$ are proper, convex, l.s.c. functions.

By **Fenchel-Legendre duality**, $F(Kx) = \max_{y \in Y} \langle Kx, y \rangle - F^*(y)$, converting the primal problem into the **saddle-point problem**:

$$\min_{x \in X} \max_{y \in Y} \langle Kx, y \rangle + G(x) - F^*(y) \quad (1)$$

Assuming:

- There exists saddle point $(\hat{x}, \hat{y})$ satisfying:

$$K\hat{x} \in \partial F^*(\hat{y}), \quad -K^*\hat{y} \in \partial G(\hat{x})$$

- F, G are "simple": resolvents $(I + \tau \partial F)^{-1}(y) = \arg\min_x \left\{ \frac{\|x-y\|^2}{2\tau} + F(x) \right\}$ and $(I + \tau \partial G)^{-1}$ have closed-form solutions or can be solved efficiently.

Main Algorithm

Initialization: Choose $\tau, \sigma > 0$ with $\tau\sigma\|K\|^2 < 1$, $\theta \in [0,1]$, $(x^0, y^0) \in X \times Y$, and set $\bar{x}^0 = x^0$.

For each iteration $n \geq 0$: (2)

1. **Dual Proximal Ascent Step (y^{n+1}):**
   - Forward gradient ascent along the coupling term $+\sigma K\bar{x}^n$ to maximize with respect to y via proximal minimization.

$$y^{n+1} = (I + \sigma\partial F^*)^{-1}(y^n + \sigma K\bar{x}^n) = \arg\min_{y \in Y}\left\{\frac{\|y - (y^n + \sigma K\bar{x}^n)\|^2}{2\sigma} + F^*(y)\right\}$$

2. **Primal Proximal Descent Step (x^{n+1}):**
   - Backward gradient descent along the adjoint direction $-\tau K^* y^{n+1}$ to minimize with respect to x via proximal minimization.

$$x^{n+1} = (I + \tau\partial G)^{-1}\left(x^n - \tau K^* y^{n+1}\right) = \arg\min_{x \in X}\left\{\frac{\|x - (x^n - \tau K^* y^{n+1})\|^2}{2\tau} + G(x)\right\}$$

3. **Linear Extrapolation ($\bar{x}^{n+1}$):**

$$\bar{x}^{n+1} = x^{n+1} + \theta(x^{n+1} - x^n)$$

Acts as an approximate extragradient step along the primal velocity direction to anticipate the next position, algebraically canceling the cross-coupling error without an implicit subproblem.

Central Theorem

Let $L = \|K\|$ and assume problem (1) has a saddle-point $(\hat{x}, \hat{y})$. Choose $\theta = 1$, $\tau\sigma L^2 < 1$, and let $(x_n, \bar{x}_n, y_n)$ be defined by (2). Then:

(a) For any n,

$$\frac{\|y^n - \hat{y}\|^2}{2\sigma} + \frac{\|x^n - \hat{x}\|^2}{2\tau} \leq C\left(\frac{\|y^0 - \hat{y}\|^2}{2\sigma} + \frac{\|x^0 - \hat{x}\|^2}{2\tau}\right),$$

where the constant $C \leq (1 - \tau\sigma L^2)^{-1}$.

(b) If we let

$$x_N = \frac{1}{N}\sum_{n=1}^{N} x^n, \qquad y_N = \frac{1}{N}\sum_{n=1}^{N} y^n,$$

for any bounded $B_1 \times B_2 \subset X \times Y$, the restricted gap has the following bound:

$$\mathcal{G}_{B_1 \times B_2}(x_N, y_N) \leq \frac{D(B_1, B_2)}{N},$$

where

$$\mathcal{G}_{B_1 \times B_2}(x, y) = \max_{y' \in B_2}\left\{\langle y', Kx\rangle - F^*(y') + G(x)\right\} - \min_{x' \in B_1}\left\{\langle y, Kx'\rangle - F^*(y) + G(x')\right\}.$$

and

$$D(B_1, B_2) = \sup_{(x,y) \in B_1 \times B_2}\left\{\frac{\|x - x^0\|^2}{2\tau} + \frac{\|y - y^0\|^2}{2\sigma}\right\}.$$

Moreover, the weak cluster points of (x_N, y_N) are saddle-points of (2);

(c) If the dimension of the spaces X and Y is finite, then there exists a saddle-point (x^*, y^*) such that

$$x^n \to x^* \qquad \text{and} \qquad y^n \to y^*.$$

Implementation: Total Variation L^1 Denoising

We consider the **TV-L^1** model, a non-smooth optimization problem.

TV-L^1 Model

$$\min_u \|\nabla u\|_1 + \lambda \|u - g\|_1$$

$$\approx$$

$$\min_u \max_p \{-\langle u, \operatorname{div} p\rangle + \lambda \|u - g\|_1 - \delta_P(p)\}$$

$$\lambda = 1.5$$

where

$u \in \mathbb{R}^{M\times N}$	:	Restored image,
$g \in \mathbb{R}^{M\times N}$	:	Noisy input image,
λ	:	Regularization parameter,
$p \in \mathbb{R}^{2\times M\times N}$	:	Dual variable,
$\operatorname{div} p$	:	Divergence of p,
$\delta_P(p)$	:	Indicator $(p \in P)$.

Method

The primal-dual algorithm (2) [**ALG1**]

$$\tau = 0.02, \qquad \tau\sigma L^2 = 1, \qquad \theta = 1 \qquad L = \|\nabla\|$$

Experimental Setup

We take an image with

$$M = 500, \qquad N = 375,$$

where the noisy input g is corrupted with 25% salt-and-pepper noise.

Ground-truth method

A well-performing method run over $\sim 10^5$ iters.

Stopping Criterion

$$\frac{E_n - E^*}{E^*} < \epsilon \qquad \epsilon \in \{10^{-4}, 10^{-5}\}$$

where E_n and E^* are the primal energy of the current iterate u_n and of the reference.

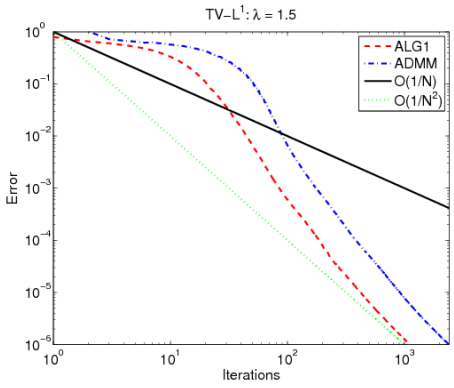
Implementation

All algorithms implemented in Matlab and executed on a 2.66 GHz CPU, running a 64 Bit Linux system.

[1] The experiment uses the boundary case $\tau\sigma L^2 = 1$. Remark 1 notes that at equality the control of the estimate for y^N is lost, while an estimate for x^N remains.

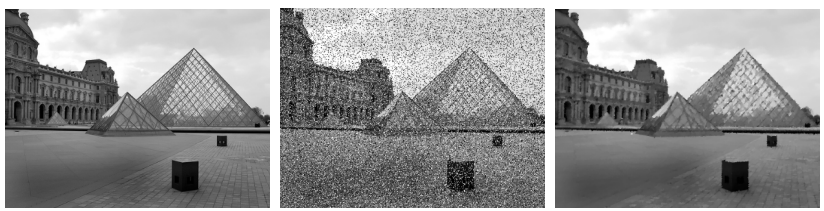
Experiment Results: $TV\text{-}L^1$ Denoising

Convergence Rate



Relative Error $(E^n - E^*)/E^*$ vs. Iterations

Visual Evaluation



(a) Clean (b) 25% S&P (c) ALG1

Quantitative Performance

Method	Iterations $\varepsilon=10^{-4}$	Iterations $\varepsilon=10^{-5}$	∇/div ($\times 10^5$)	CPU (s) ($\times 10^5$)
ALG1	**187**	**421**	**842**	**36.0**
ADMM	385	916	—	80.0
EGRAD	2462	8736	34,944	1360.0
NEST	2406	15538	—	1387.0

**ADMM: Alternating dir. Method ; EGRAD: Extra-Gradient ; NEST: Nesterov*

Key Takeaways

- **Wall-Clock Speedup: ALG1** is 2.2× **faster** than state-of-the-art ADMM.
- **Convergence Rate:** Follows theoretical $O(1/N)$, with empirical tail convergence closer to $O(1/N^2)$.